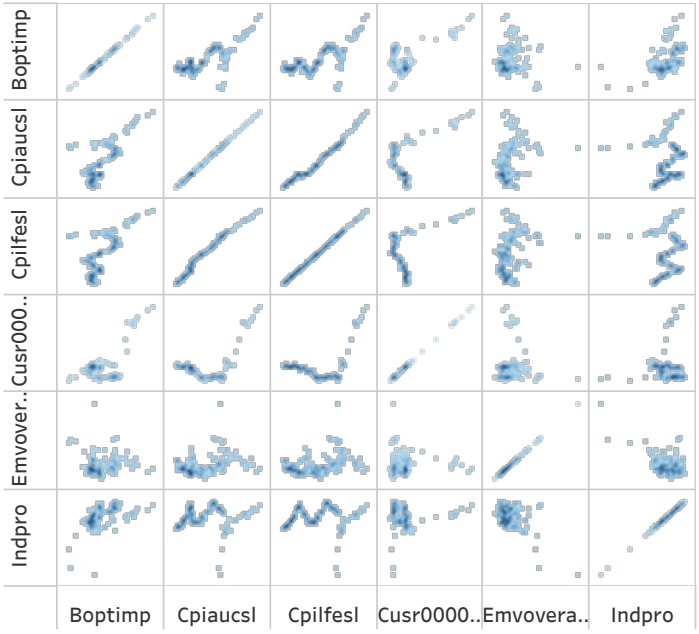
FRED Data Correlation

Searching for correlation using visualization

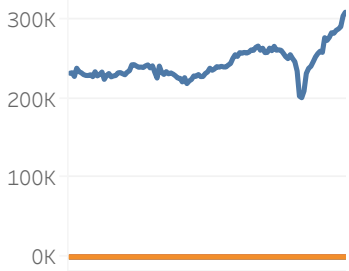


Created by Paul Larison

A Graph of Graphs

Turning all the fields into scatter plots lets us look for patterns. The intersections of CPIAUSL and CPILFESL look promising.

Raw Data

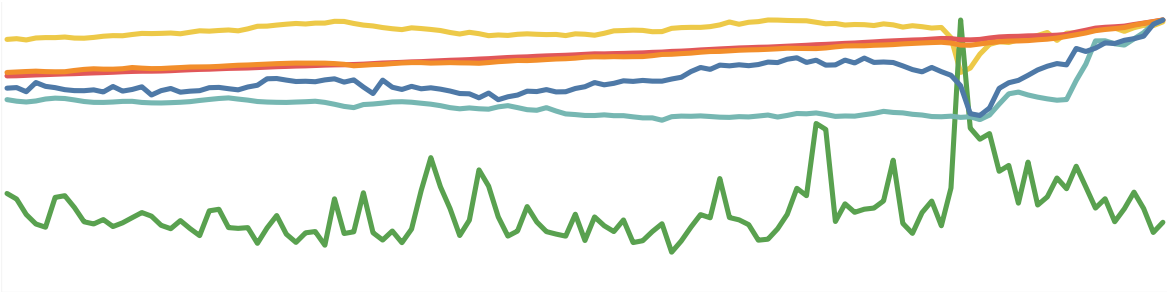


Using the Raw Data to create a line chart doesn't go well. The high scale of BOPTIMP renders the other fields featureless.

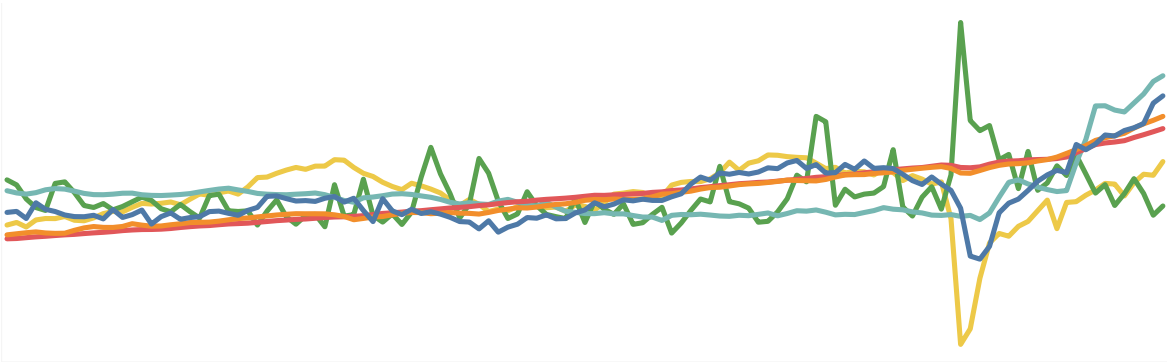
Normalization

is a process that re-scales values to fit them to the same numeric range without changing the ratios. Two ways to do this are the min-max method and the z-score method, which give slightly different results. Once the tables are normalized, we can see all the fields on the same graph.

Normalized, Min-Max



Normalized, Z Score



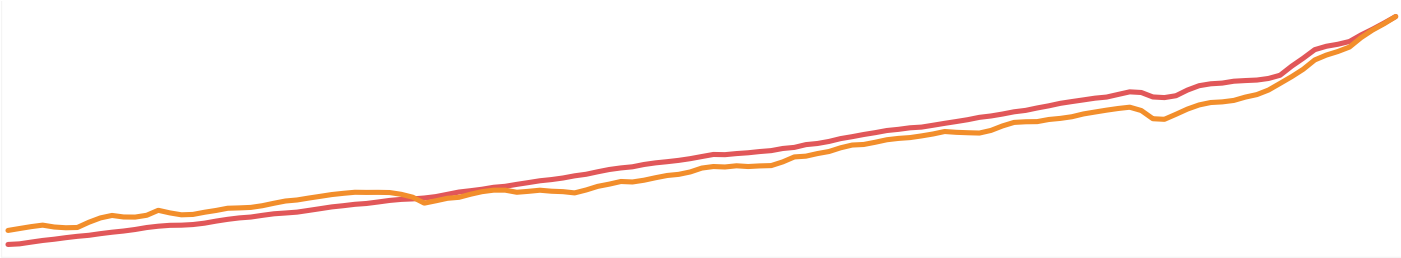
- BOPTIMP
- CPIAUCSL
- CPILFESL
- CUSR0000SETA02
- EMVOVERALLEMV
- INDPRO

Results

CPILFESL and CPIAUCSL

This is by far the strongest correlation in the data and has been apparent in every view we've seen.

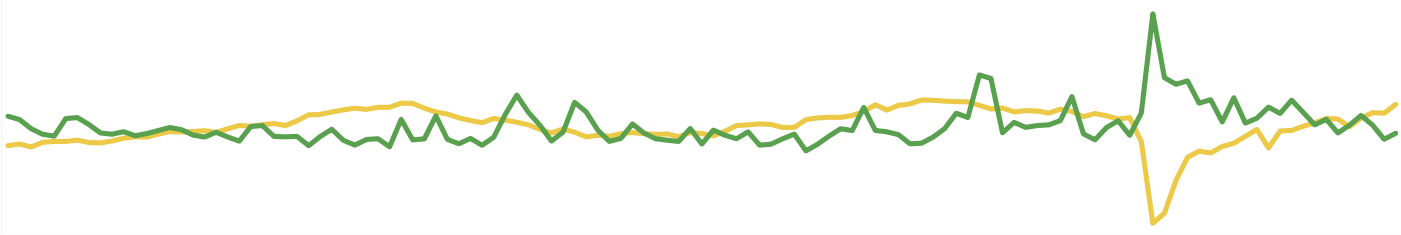
Pearson's Coefficient: 0.99



EMV and INDPRO

The opposite peaks in the z-score graph make it look like this might be a negative correlation. Once isolated, it becomes apparent that the correlation isn't very strong.

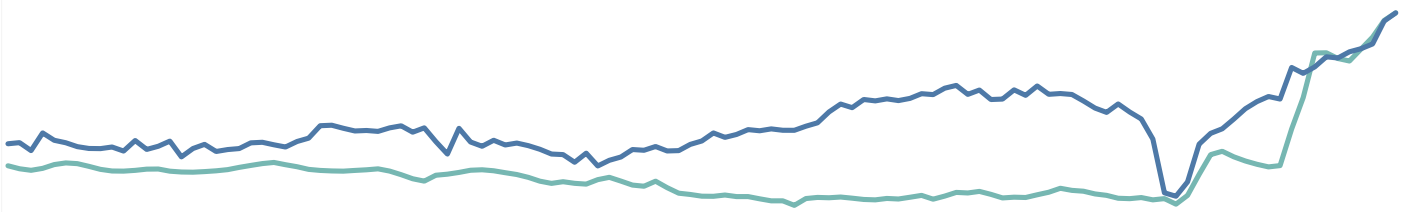
Pearson's Coefficient: -0.53



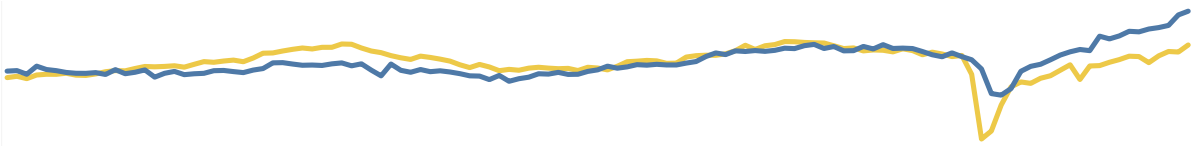
CUSR and BOPTIMP

The smaller movements in this pair track together fairly well, although the lines drifted apart in the middle of the graph.

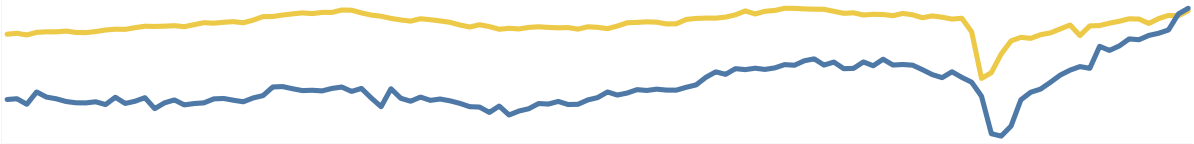
Pearson's Coefficient: 0.58



BOPTIMP and INDPRO, Z-Score



BOPTIMP and INDPRO, Min-Max



BOPTIMP and INDPRO look more correlated under z-score normalization than they do with min-max. The actual correlation is moderate, with a Pearson's coefficient of 0.52